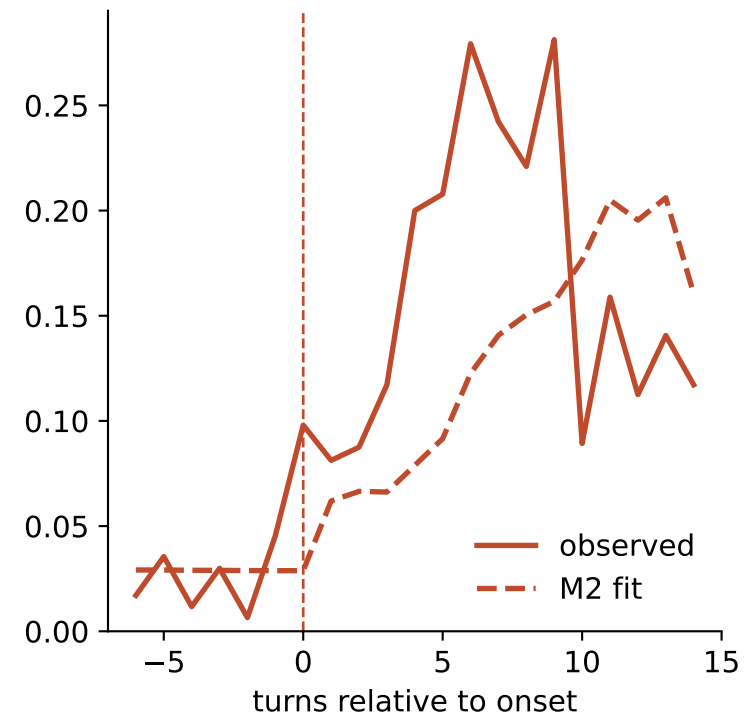
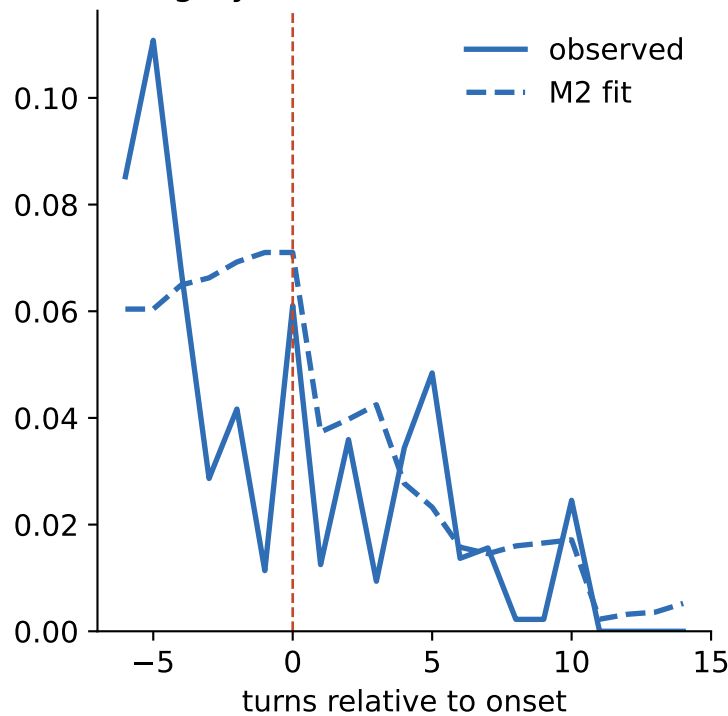
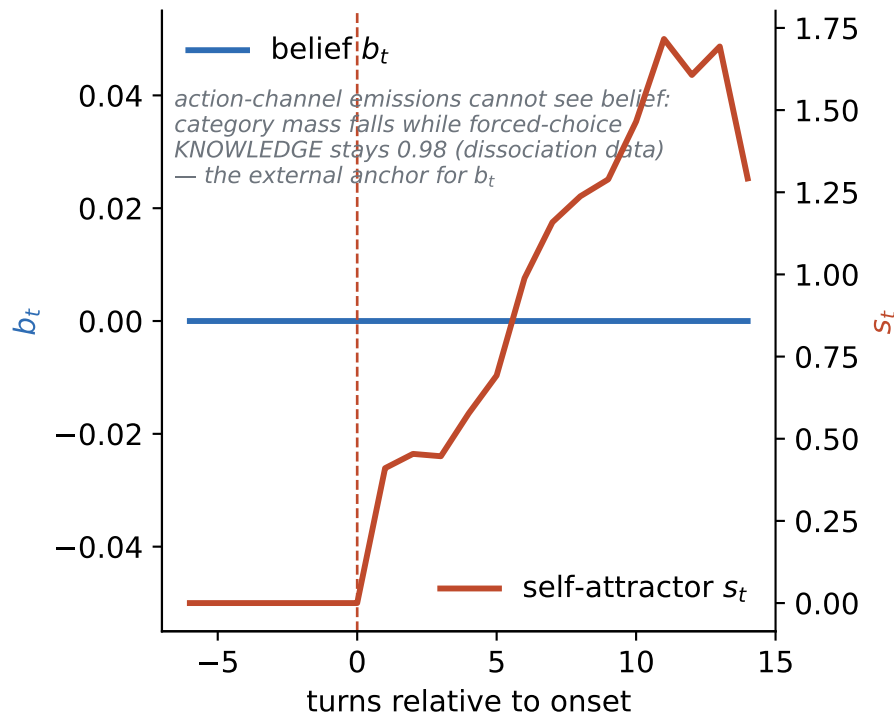


Two-process state-space fit (Qwen3-32B strict city, 24 games)

accumulates; b_t is UNIDENTIFIED from actions alone ($\eta_b \rightarrow 0$)



Model comparison and coefficients

M2 two-process: NLL 23336.1 AIC 46740.1 6-fold held-out NLL 24957.9

M1 belief-only: NLL 26083.7 AIC 52227.4 6-fold held-out NLL 27747.2

$\Delta\text{AIC (M1-M2)} = 5487.3$ $\Delta\text{held-out (M1-M2)} = 2789.4$ (positive favors two-process)

M2 parameters: $\eta_b=0.000$ $\rho=0.884$ (persistence) $\eta_s=0.456$

$\alpha_b=0.763$ $\alpha_s=1.305$ (category suppression) $\beta_s=2.849$ (stem drive) $\gamma_s=2.249$ (repeat drive)

Mixed-effects AR: $\text{logit}(p_{\text{stem}}_{t+1}) \sim p_{\text{stem}}_t + \text{sim}(a_t) + p_{\text{cat}}_t + (1|\text{game})$

Intercept: $\beta=-4.306$ (se 0.110, $p=0$)

p_{stem} : $\beta=+3.860$ (se 0.527, $p=2.5e-13$)

sim: $\beta=+1.360$ (se 0.204, $p=2.4e-11$)

p_{cat} : $\beta=-0.650$ (se 0.391, $p=0.096$)

Causal anchors from experiments (not re-fit):

sim/ β_2 : transplant – 3 planted family words → stem mass 0.57 vs 0.00/0.01 (onset 8/8)

attribution/ β_4 : seed matrix – list 0.53 / self 0.24 / partner 0.13 / other 0.09

persistence: dose-4 hysteresis – one forced unrelated word: 0.41 → 0.36